# Can Frontier LLMs One-Shot Translate Lakota? (Mostly No.)

*Evaluating Frontier LLM Translation Capability for Lakota*

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## No frontier LLM reliably translates *Lakȟótiyapi*.

Even measuring how badly they fail is hard.

## Why test this?

*Lakȟótiyapi* (Lakota) is a Siouan language UNESCO classifies as **critically endangered** — fewer than 2,000 first-language speakers, most over 65.

Today's LLMs list translation support across dozens of languages, but providers don't say what they can or can't do per language. A reasonable user can't tell Lakota apart from any of the others on the list — *if it does all these, why not this one?* Direct evaluation is the only way to find out.

## Is it just memorized?

The evaluation set comes from a published dictionary, so contamination is the obvious worry. The Lakota source appears verbatim online for ~10% of pairs, but the Lakota and its English reference together for only 0.5%. Scores spread widely (6–61% equivalence), not clustered at ceiling — the signature of partial capability, not recall.

## Setup

- **7 models.** Proprietary: Gemini 3.1 Pro, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2. Open-weight: DeepSeek-V4-Pro, Qwen3.6-Plus, GLM-5.1.
- **200 sentence pairs** from the New Lakota Dictionary, conversational register, **both directions.**
- **Two conditions:** baseline vs. extended reasoning, where the API permits.
- **Metrics:** chrF++; diacritic-normalized chrF++; an LLM *semantic* judge on the English side.

## LLM-as-judge for L → E

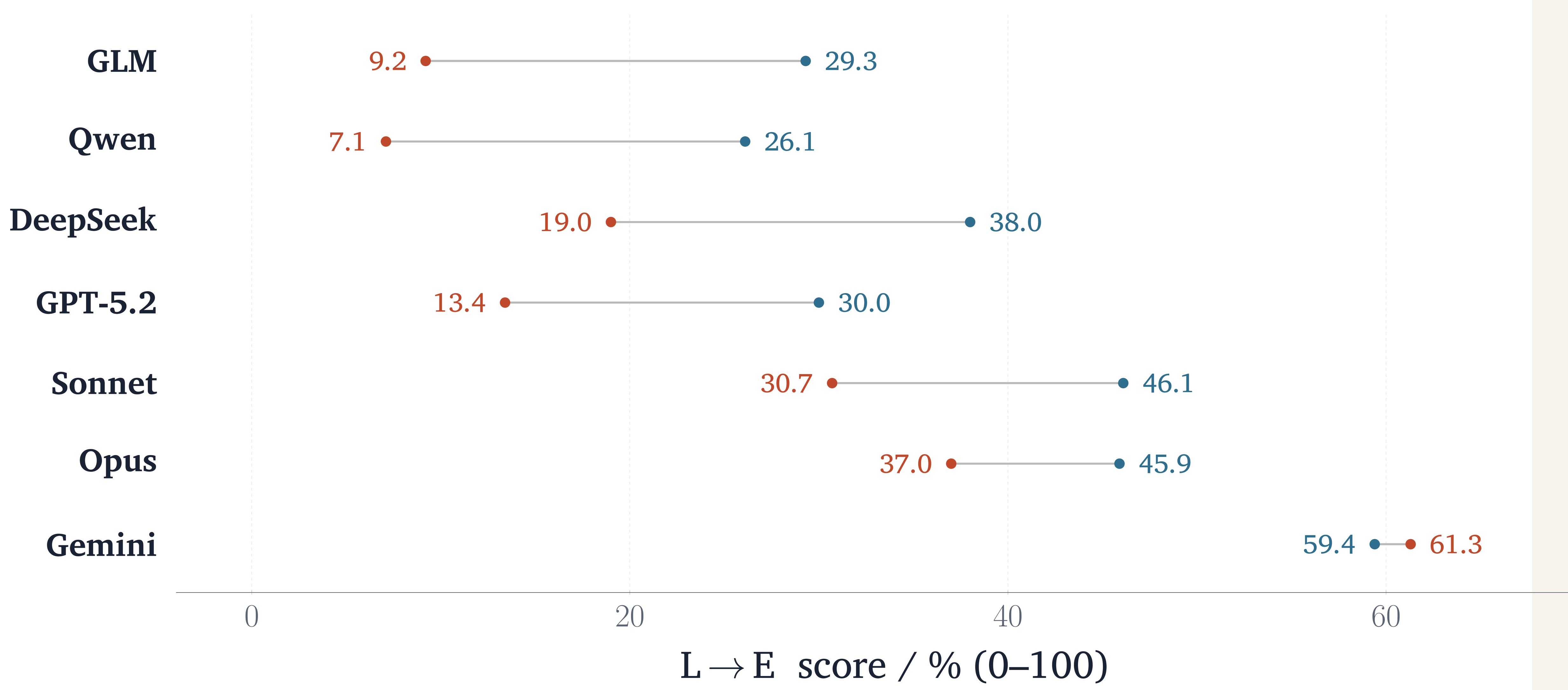
chrF++ measures character and word overlap, but it can reward plausible nonsense. We wanted an automated measure that could take meaning into account. We use an LLM-as-judge: for the L → E direction the output is English, so a model can assess whether each translation carries the reference's meaning. To keep the judge from favoring its own family, every pair is scored by two LLMs from different families, Gemini 3 Flash and an independent Llama-3.3-70B. They agree substantially (Cohen's $\kappa = 0.75$, 0.89 quadratic-weighted), so the judgments are cross-validated rather than one model's unchecked read.

## Standard Lakota Orthography

č š ž ǧ ȟ   á é í ó ú   ŋ   ʼ

SLO encodes phonemic distinctions with diacritics. To account for variance in orthographic conventions, we re-score with the marks stripped: the gain is only +3.3–6.5 chrF++ (E → L), small enough to suggest the marks aren't the main error these models make.

## Surface metrics overstate capability




Each row is one model, ranked by gap size (largest at top). Connector length is how far the surface score chrF++ (steel) overstates % meaning-equivalent (terracotta). Only Gemini's two scores nearly agree.

## Translation quality (chrF++)

| Model | L → E | E → L |
|---|---|---|
| *Proprietary* | | |
| Gemini 3.1 Pro | **59.4** | **42.6** |
| Claude Opus 4.6 | 45.9 | 34.3 |
| Claude Sonnet 4.6 | 46.1 | 27.0 |
| GPT-5.2 | 30.0 | 23.6 |
| *Open-weight* | | |
| DeepSeek-V4-Pro | 38.0 | 29.2 |
| GLM-5.1 | 29.3 | 14.6 |
| Qwen3.6-Plus | 26.1 | 18.4 |

Best condition per model. Every model scores higher *out of* Lakota than *into* it.

## Semantic equivalence (L → E)

| Model | Equiv. | Partial | Not |
|---|---|---|---|
| Gemini 3.1 Pro | **61.3** | 23.1 | 15.6 |
| Claude Opus 4.6 | 37.0 | 21.0 | 42.0 |
| Claude Sonnet 4.6 | 30.7 | 23.8 | 45.5 |
| DeepSeek-V4-Pro | 19.0 | 14.0 | 67.0 |
| GPT-5.2 | 13.4 | 9.8 | 76.8 |
| GLM-5.1 | 9.2 | 6.5 | 84.2 |
| Qwen3.6-Plus | 7.1 | 7.6 | 85.4 |

% of judged pairs, reasoning condition.

## How models fail

| Reference | Model | Output | chrF++ |
|---|---|---|---|
| We live in a small trailer house with no electricity. | DeepSeek | "small *sacred* tipi" | 18.6 |
| Make coffee (*Wakȟálya yo*). | DeepSeek | "Sanctify it!" | 2.9 |
| Did my daughter call? | GLM | "Did my younger brother come?" | 27.6 |
| Do you have a car? | GPT-5.2 | "Do you have a horse?" | 66.4 |

Idioms and coined terms are translated *element by element*: *Wakȟálya yo* ("make coffee," literally "make it boil") becomes "sanctify it." A one-word near-miss — "car" rendered "horse" by a *frontier* model — still scores 66.4 chrF++. Surface overlap and meaning preservation diverge.

## Phrasebook concentration (L → E)

| Model | Sc. 4 | Sc. 5 | Sc. 6 | Δ |
|---|---|---|---|---|
| Gemini 3.1 Pro | 52.7 | 58.2 | **81.9** | +29.2 |
| Claude Opus 4.6 | 41.6 | 41.0 | 65.6 | +24.0 |
| Claude Sonnet 4.6 | 40.3 | 45.1 | 64.1 | +23.8 |
| DeepSeek-V4-Pro | 33.0 | 33.9 | 59.6 | +26.6 |
| GPT-5.2 | 27.9 | 23.7 | 45.7 | +17.8 |
| Qwen3.6-Plus | 23.3 | 22.1 | 41.7 | +18.4 |
| GLM-5.1 | 27.3 | 26.0 | 41.0 | +13.7 |

chrF++ stratified by NLD conversational score: 4 = more formal/complex, 6 = most conversational. Every model lifts sharply at 6 — performance concentrates on high-frequency phrasebook-like items rather than reflecting general capability.

## Takeaways

- No frontier LLM reliably translates Lakota, either direction (2026).
- chrF++ overstates low-resource capability — pair it with a meaning check.
- Open-weight doesn't close the gap.

Extended reasoning changes refusal behavior more than quality (open-weight E → L: **0** → **36** refusals when enabled).

## Origin

This evaluation grew out of a long-running interest in the possibility of crowdsourcing a Lakota training corpus. It started as a User Interface design problem: what could community-driven tools look like for gathering training data for indigenous-led AI? In a seminar, faculty gave us the provocation: that today's LLMs are good enough you could just *"vibe-code"* a Lakota translator over a weekend. I wanted to investigate if this was feasible. My background in software development and LLM evaluations drove me to create this Lakota benchmark.

**Data & code**
github.com/robotson/lakota-translation-benchmark